9a.Simulation of Ground Wave and Sky Wave Propagation using MATLAB

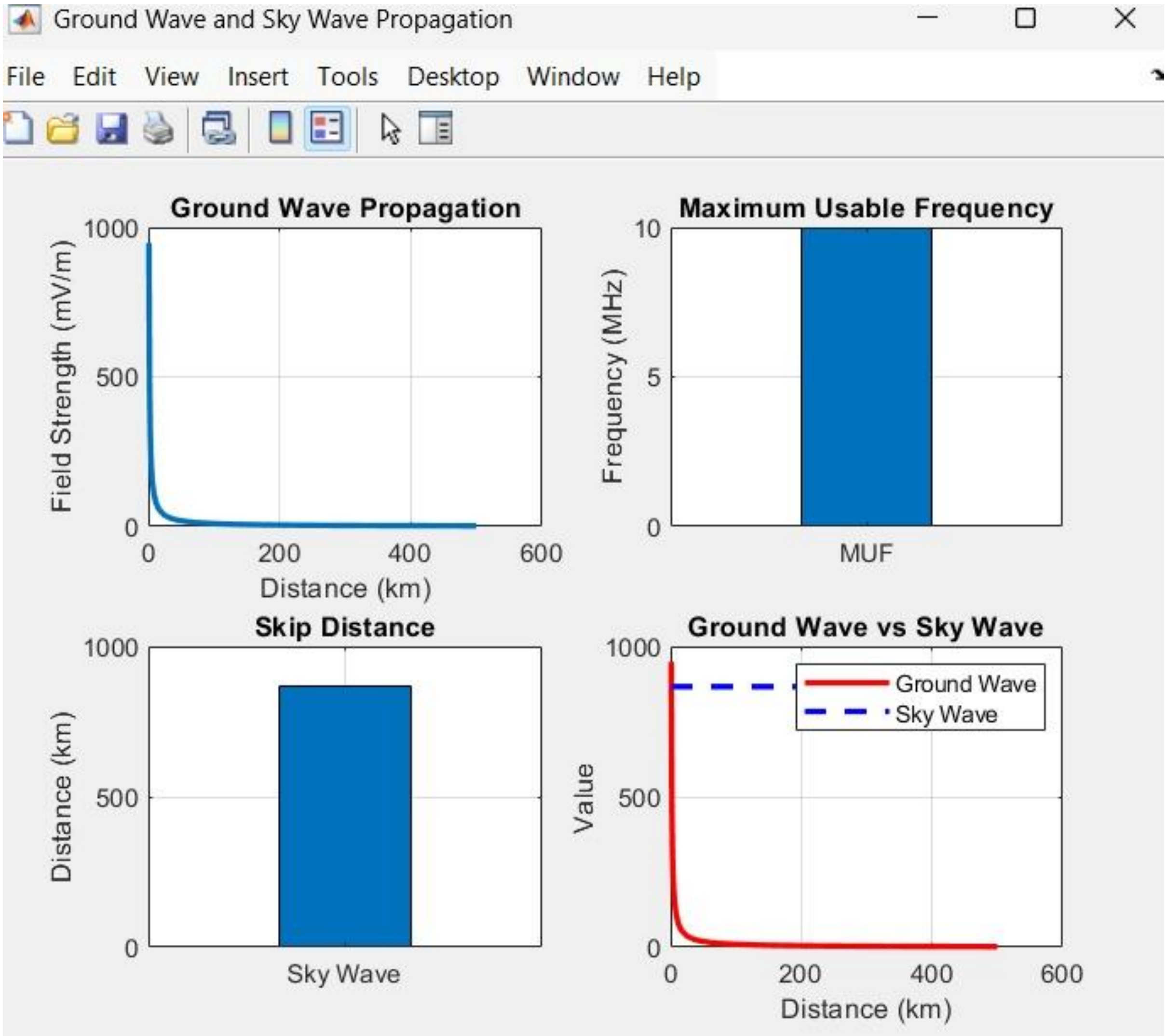
OVERALL MATLAB CODE:

```
clc;
clear;
close all;
%% Objective 1: Ground Wave Propagation
Pt = 10; % Transmitted Power (kW)
d = 1:1:500; % Distance (km)
E_ground = (300*sqrt(Pt))./d;
%% Objective 2 : Sky Wave Propagation
h = 250; % Ionosphere Height (km)
theta = 60; % Angle of Incidence (degrees)
fc = 5; % Critical Frequency (MHz)
Skip_Distance = 2*h*tand(theta);
MUF = fc*sec(deg2rad(theta));
%% Objective 3 : Comparative Analysis
Sky_Range = Skip_Distance*ones(size(d));
%% Display Results
fprintf('\nGROUND WAVE AND SKY WAVE ANALYSIS\n');
fprintf('----------------------------------\n');
fprintf('Transmitted Power = %.2f kW\n',Pt);
fprintf('Ionospheric Height = %.2f km\n',h);
fprintf('Skip Distance = %.2f km\n',Skip_Distance);
fprintf('Maximum Usable Frequency (MUF) = %.2f MHz\n',MUF);
```

```
%% Plot Results
figure('Name',...
'Ground Wave and Sky Wave Propagation',...
'NumberTitle','off');
subplot(2,2,1)
plot(d,E_ground,'LineWidth',2)
grid on

xlabel('Distance (km)')
ylabel('Field Strength (mV/m)')
title('Ground Wave Propagation')
subplot(2,2,2)
bar(MUF)
grid on
title('Maximum Usable Frequency')
ylabel('Frequency (MHz)')
set(gca,'XTickLabel',{'MUF'})
subplot(2,2,3)
bar(Skip_Distance)
grid on
title('Skip Distance')
ylabel('Distance (km)')
set(gca,'XTickLabel',{'Sky Wave'})
subplot(2,2,4)
plot(d,E)
```

OUTPUT:


Ground Wave and Sky Wave Propagation
File Edit View Insert Tools Desktop Window Help
Ground Wave Propagation
Field Strength (mV/m)
1000
500
0
0
200
400
600
Distance (km)
Maximum Usable Frequency
Frequency (MHz)
10
5
0
MUF
Skip Distance
Distance (km)
1000
500
0
Sky Wave
Ground Wave vs Sky Wave
Value
1000
500
0
Ground Wave
Sky Wave
0
200
400
600
Distance (km)